

AIDAN M. MOKALLA

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UNDERGRADUATE EDUCATION _____ GPA: 3.6/4

- Reed College**, Portland, Oregon, USA, 2025
- + Spanish minor, sole member of graduating class with double majors: Computer Science & Linguistics,
 - + A grades in *Real Analysis*, *Linear Algebra*, *Deep Learning*, *Linguistic Field Methods*, and *Psycholinguistics*,
- Bryn Mawr College**, Dushanbe, Tajikistan, 2024
- + Fellowship funded by international non-profit *American Councils* to research Persian language varieties,
- La Universidad de Carlos III** and **Middlebury College**, Madrid, Spain, 2023
- + Spanish-language courses including *Artificial Intelligence*, *Data Science*, and *International IP Law*.

SELECTED PUBLICATION _____

- Source of the Storm: Interpreting Weather Models to Detect Cyclogenesis**, *forthcoming*
- + Applying post-superposition statistical interpretation methods to weather prediction models to improve localization of rare climatic events and demonstrate methodological generalizability beyond transformer-based models,
- Evaluating Diffusion Language Models Beyond Benchmarks**, *COLM 2026*
- + Found that people, when evaluating the outputs of the recently invented *diffusion-based* LLM architecture, consistently over-trust AI outputs at a significantly lower rate than when evaluating outputs of traditional LLMs,
- Interpretable Signals for VLA Model Failure**, *ICML 2026*
- + Discovered representational mechanistic predictors of vision-language-action model task failure,
- Fault- and Network-Agnostic Agreement**, *with Erica Blum*, *TCC 2026*
- + Proved tight running-time bounds for all algorithms achieving network-agnostic Byzantine consensus,
- Language-Agnostic Concept Spaces Found in Large Language Models**, *with Greg Anderson*, *thesis, 2025*
- + Designed, implemented, and evaluated novel architecture to interpret transformer-based LLMs by leveraging natural-language translation of ground-truth semantics. My approach learns convergent “concept spaces” over LLM activations in parallel, yielding more consistent LLM interpretations than prior work,
- Modulated Signal Filtering: Wouldn't it be Noise?**, *with Sameer ud Dowla Khan*, *thesis, 2025*
- + Collected over 3,000 participant responses to investigate how listeners use unique cognitive strategies to parse spoken speech versus to parse song lyrics, using both deep and symbolic statistical methods,
 - + Demonstrated that we use contextually-differentiated mechanisms to filter noise from the “linguistic signal” by making significantly more accurate predictions of how people interpret unclear speech under this prior.

LEADERSHIP & EXPERIENCE _____

- Algoverse AI Research**, *Machine Learning Researcher, Fellow, & Mentor* *September 2025 – present*
- + Investigating and publishing methods of explainable AI and climate science,
- Linguistic Diversity Initiative**, *Lead Volunteer Ambassador*, *June 2023 – August 2025*
- + Leading a team of three translators, I compiled and published dictionaries of terminology used in Portland Public’s science curricula, translated into 6 languages, for households with low English proficiency, already enabling over 800 families to engage in their students’ education,
 - + Taught seminars to students and faculty about sociolinguistic research on language use in undergraduate settings,
 - + Organized and lead twice-monthly meetings of over 50 volunteers and larger events,
- Reed College Student Body**, *Senator & Honor Councillor*, *September 2021 – May 2025*
- + Peer-elected to serve in faculty committees including the *Committee on Academic Policies and Planning* and *Committee on Computer Policy and Planning*, disburse 6-figure Student Body Fund and other resources to college organizations, and interview and hire all student applicants for on-campus roles,
 - + Mediated and resolved conflicts between students, professors, and other members of the college community,
- Stoel Rives, LLC**, *Intern Practise Assistant*, *May – August 2022*
- + Implemented compliant automation of docketing, court appearances, and other events from firm-wide databases,
 - + Maintained communication on behalf of attorneys while drafting and proofreading attorney correspondences.

SKILLS & ACCOLADES _____

- Computational**: Python, various data and NLP pipelines, PyTorch, NumPy, JS, Next.js, C++, PHP, SQL, Swift,
- Awards**: Presidential Commendation for Academic Excellence, Reed College Music Scholarship, National Diversity in STEM Grant, Department of State Critical Languages Scholarship,
- Certifications**: Federally eligible for non-competitive civil service, licensed in Conflict Mediation & Resolution, CPR,
- Languages**: Fluent Spanish, English, intermediate Farsi, Tajik, French, basic Portuguese, Dari, Chinese (Mandarin).